

How Lidar Is Enabling AUTONOMOUS DRONES

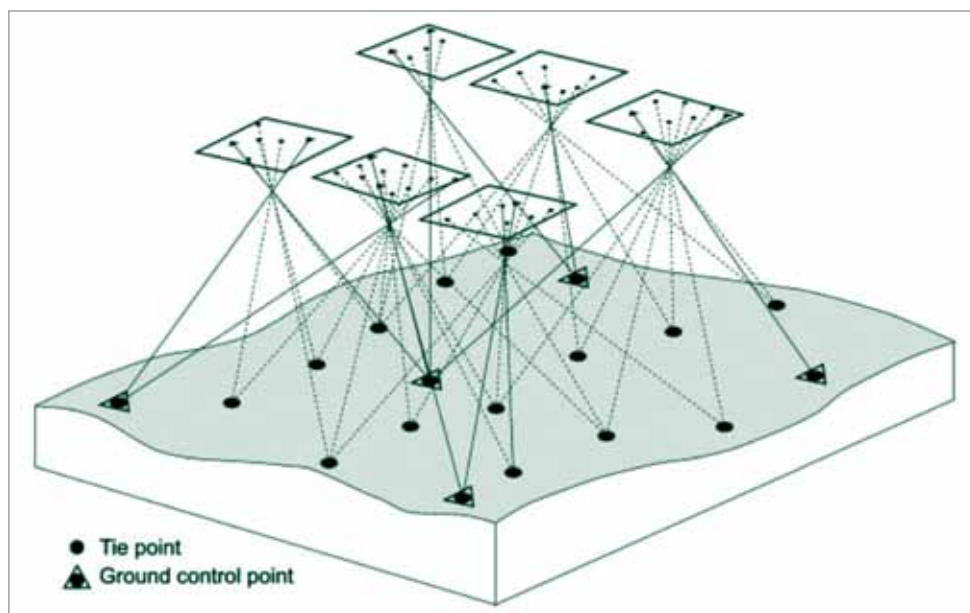
Lidar, which stands for light detection and ranging, is a remote sensing method that uses light in the form of a pulsed laser to measure variable distances to the Earth. These light pulses—combined with other data recorded by the airborne system—generate precise, three-dimensional information about the shape of the Earth and its surface characteristics. The information can be highly useful, as described below



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Photogrammetry in image scanning and re-construction using drones (Credit: www.scoutaerial.com.au)

An autonomous drone is known by different names, such as unmanned aerial vehicle (UAV), unmanned aircraft system (UAS), or remotely piloted aircraft system (RPAS). But, in terms of industrial application, they are used to monitor the entire region or terrain inside or outside industrial units, which sometimes becomes quite cumbersome for humans to perform due to hazardous settings or environmental instability.

To do such monitoring, innovative platforms are being developed across the globe. Such platforms provide complete

end-to-end, autonomous solutions for collecting aerial data. An industrial UAV is like any other drone having IoT web capability, communicating directly with industrial control centre and all the (IoT) sensors for temperature, humidity, atmospheric pressure, gas, motion, electric and magnetic field, high-resolution full HD camera with object detection and tracking to carry out the required operations.

An autonomous drone (aka drone-in-a-box) gathers all sorts of data during its flight path, or customised data requested from the ground station, and sends it to the monitoring station. Without a ground